developed by director Yutaka Saito of Seaman Artificial Intelligence Laboratory. The uniqueness of this engine is its memory and personality.

Smart assistants and smartphone AI assistants so far have only answered questions one by one and can respond to questions and instructions quickly, but they do not respond to calls that are not in their library. However, Saito's AI receives words, sorts out the context up to that point, returns new questions and even gives advice.

AI with personality may not always be an excellent assistant, but warm communication will shake people's emotions. BOCCO language is an electronic sound, so it will not be like a conversation between people like a demo, but the human side may complement the meaning and feel more sympathetically. The AI conversation engine is still under development.

—Deepshikha Shukla

5 A BRAIN ACTIVITY MEASURING DEVICE

Brain function is known to decrease with age. But just like muscle-strength training, our brain too can maintain and improve brain function regardless of age with proper training.

HOT-2000 XB-01

Brain training can now be performed by anyone, anywhere and anytime—from maintaining the mental health of the elderly to increasing brain power of professionals who want to improve productivity and school students who want to improve their grades. It can help increase processing speed, memory, decision-making and more.

NeU is a brain science company, founded in 2017 by fusing the cognitive brain science discoveries of Tohoku University with portable brain measurement technologies of Hitachi High Technologies Corp. The company has developed compact and wearable optical topography devices using near infrared spectroscopy (NIRS) technology to measure brain activity in a more user-friendly environment—the devices are not meant for medical use.

XB-01 is an ultra-compact brain activity measuring device weighing thirty grams and measuring 80mm x 40mm x 13mm. It can transfer brain activity data via Bluetooth in real time to any smartphone or tablet. XB-01 includes a built-in battery that lasts up to three hours, USB charging terminal, easy-to-use buttons and switches, a handy holder and designated headband. Brain activity is measured using NIRS technology, and the brain's rate of blood flow change is measured using a weak NIR light.

Six-axis acceleration sensors are built into the main body to detect the head's movement and body movement noise. Combining multiple parameters in addition to measuring brain activity allows XB-01 to create measurement indexes with improved accuracy. Its real-time measurement provides extensive usage in various scenarios, such as movement and cognitive function training, stress coping training, learning and much more.

The device can be used inside the hat of a transport driver or a worksite helmet and so forth, to help prevent accidents and improve work productivity. It enables visualising brain activity during the learning process, thus allowing the learner to objectively know the degree of his/her concentration level. The learner can then modify and make changes to the environment to better enhance concentration and memory abilities.

HOT-2000 is another portable brain activity measurement device that can be used in everyday scenarios. It has the capability to simultaneously measure cerebral blood flow change, heart rate and head acceleration in real time. It can also transmit brain activity data wirelessly through Bluetooth, to be recorded on smartphones and tablets. It is portable and lightweight, weighing only 129 grams. It can start brain activity measurement immediately, right after the individual wears the device.

HOT-2000's sensor portion features a slide mechanism that can be adjusted in the range of 60mm left-to-right. The device can be used continuously for around four hours. With six-axis acceleration sensors, detailed movement of the head can be acquired to analyse body movements. It is intended for use by companies and universities for research purposes. It is compatible with Android OS version 7.0 or higher. It has application software, Measure 3.0 and Analysis 3.0 to analyse brain activity data, including noise processing, blocking and baseline correction. It also edits and saves the measurement design function.

NeU and FOVE have developed an integrated virtual reality (VR) device that can simultaneously acquire brain activity to visualise and quantify a user's interest, concentration and attention in the virtual realm. An objective and quick analysis of data acquired from purchasing and consuming activities of multiple users ensures better products and services. EFY

—Deepshikha Shukla